



Cognition and Instruction: Their Historic Meeting Within Educational Psychology

Richard E. Mayer
University of California, Santa Barbara

The purpose of this article is to examine the emergence of cognitive approaches to instruction, with hope that an understanding of the origins of this field will yield perspective on where it is going. First, an historical analysis of the relation between psychology and education, especially cognitive psychology and instruction, is undertaken. Then, an historical overview is provided of the progression of 3 views of learning and instruction: learning as response acquisition, learning as knowledge acquisition, and learning as knowledge construction. Next, cognitive theory is examined in more detail, with a particular focus on cognitive processes involved in learning. Finally, examples of progress and potential progress in the area of cognition and instruction are provided, through a focus on the emergence of psychologies of subject matter and on the cognitive analysis of individual learning strategies.

The Science of Psychology and the Practice of Education: A Vision of Educational Psychology

I begin with a vision of how the practice of education could be improved by the science of psychology:

The efficiency of any profession depends in large measure upon the degree to which it becomes scientific. The profession of teaching will improve (1) in proportion as its members direct their work by the scientific spirit and methods, that is by honest, open-minded consideration of facts, by freedom from superstitions, fancies or unverified guesses, and (2) in proportion as the leaders in education direct their choices of methods by the results of scientific investigation rather than by general opinion. (Thorndike, 1906, p. 257)

This quote from E. L. Thorndike's *The Principles of Teaching Based on Psychology* exemplifies that at the opening of this century, confidence that science could improve practice in the field of education was high.

For 100 years, psychologists and educators have recognized the potential relation between their fields. As early as 1892, William James (1899/1958) presented his famous lectures on psychology applied to education and later published them as

Talks to Teachers. As it is today, the field of education was in ferment, with educators searching for reforms that would bring respect to the teaching profession. Although he recognized the potential value of psychology for education, James (1899/1958) was somewhat hesitant in making specific recommendations:

The desire of the schoolteachers for a completer professional training, and their aspiration toward the professional spirit in their work, have led more and more to turn to us for light on fundamental principles. And in the few hours which we are to spend together you look to me, I am sure, for information concerning the mind's operation, which may enable you to labor more easily and effectively in the several classrooms over which you preside. Far be it from me to disclaim for psychology all title to such hopes. Psychology ought certainly to give the teacher radical help. And yet I confess that, acquainted as I am with the height of your expectations, I feel a little anxious lest, at the end of these simple talks of mine, not a few of you may experience some disappointment at the net results. (p. 22)

James acknowledged two difficulties in applying psychology to education. First, he noted that psychology's description of "the elements of the mental machine . . . and their workings" (James, 1899/1958, p. 26) does not translate directly into specific educational prescriptions—foreshadowing the modern distinction between descriptive learning theory and prescriptive instructional theory (Bruner, 1964). James (1899/1958) warned teachers: "You make a great, a very great mistake, if you think that psychology, being a science of the mind's laws, is something from which you can deduce definite programmes and schemes and methods of instruction for immediate schoolroom use" (p. 23). It is unrealistic for teachers to expect simple, step-by-step procedures guaranteed to work for all children in all situations, and it is unethical for psychologists to profess to be able to supply them.

Second, James (1899/1958) correctly noted that in 1892 psychology lacked the necessary database:

This article is based partially on an unpublished paper entitled *Cognitive Theory for Training and Education*, which was presented in August 1991 at the Mini-Convention on the Contributions of Psychology to Learning and Education in San Francisco.

The description of cognitive theory presented in the section of this article entitled "Three Views of Learning and Instruction" and in Figure 1 was first discussed in the description presented in "Guiding Students' Cognitive Processing of Scientific Information" (Mayer, 1992a).

Correspondence concerning this article should be addressed to Richard E. Mayer, Department of Psychology, University of California, Santa Barbara, California 93106.

We have been experiencing something like a boom in psychology in this country. Laboratories and professorships have been founded, and reviews established. . . . 'The new psychology' has thus become a term to conjure up portentous ideas. . . . In my humble opinion there is no 'new psychology' worthy of the name. There is nothing but the old psychology which began in Locke's time, plus a little physiology . . . and theory of evolution, and few refinements of introspective detail, for the most part without adaptation to the teacher's use. (pp. 22-23)

As have other educational psychologists throughout this century, I have been haunted by the words of James (1899/1958) and Thorndike (1906) concerning the relation between psychology and education. While sharing their 100-year-old dream that the "proper application of psychological findings might lead the way to better instruction in all schools" (Woodring, 1985, p. 6), I also share their concerns about the appropriate relation between science and practice.

Educational Psychology as a Guiding Science

At the opening of this century, educational psychology began as what Cubberly (1920) called a "guiding science of the school" (p. 755), a discipline the content of which was defined by the problems of education and the methods of which were defined by the science of psychology. In his preface to the first issue of the *Journal of Educational Psychology*, Thorndike (1910) proclaimed that educational psychology would apply the methods of exact science to the problems of education (see also Levin, 1992; Walberg & Haertel, 1992).

Despite the high hopes of early educational psychologists, the discipline was in disarray by mid-century. Grinder (1989) summarized three ways in which educational psychology lost its mantle as a guiding science for education: (a) through withdrawal: educational psychologists failed to accept responsibility for contributing to educational policy; (b) through fractionation: educational psychologists failed to achieve a coherent theoretical perspective; and (c) through irrelevance: educational psychologists failed to study practical educational problems in natural settings.

As the end of the 20th century approaches, the problems of education remain, but the science of educational psychology has begun to overcome many of the problems cited by Grinder: Withdrawal is giving way to participation in public policy debates, fractionation is giving way to a unified cognitive approach,¹ and irrelevance is giving way to the study of how real students learn real educational subject matter.

First, educational psychologists have become increasingly involved in educational policy issues. For example, the August 1991 American Psychological Association-sponsored "Mini-Convention on the Contributions of Psychology to Learning and Education" held in San Francisco, reflects a growing commitment by psychologists to participate in the educational reform movement. The six national educational goals (see U.S. Department of Education, Office of Educational Research and Improvement, 1991) currently under discussion include objectives that are of interest to educational psychologists, such as ensuring that students achieve "competency over challenging subject matter including English, mathematics, science, history, [and] geography" and that students de-

velop "the ability to reason, solve problems, apply knowledge, and write and communicate effectively" (pp. 3-4).

Second, the behaviorist movement, which dominated psychology until the 1950s, stifled much useful research in the discipline. In contrast, educational psychology has experienced what Scandura et al. (1981) called "the paradigm shift from S-R [stimulus-response] to information processing" (p. 367) and what Di Vesta (1989) recognized as "the recent era of transition from behavioral to cognitive psychology" (p. 39). Instead of focusing on the role of reinforcers in creating behavior, cognitive psychology has "brought about a renewal of concern about the learner's role in instruction" (Di Vesta, 1989, p. 39). In particular, "an important aspect of the cognitive movement in education was the conscious recognition of the learner as an active participant in the learning process." (Di Vesta, 1989, p. 54).

Third, educational psychology has moved from the study of behavior on artificial tasks in sterile laboratory environments to the study of the cognitive processes of students as they engage in school tasks in more natural settings. For example, instead of studying rote memorization of lists of nonsense syllables, researchers in educational psychology now investigate how to improve the understandability of school textbooks; instead of measuring the rate at which rats run through mazes, researchers now evaluate the strategies that children use on school tasks; and instead of studying how adults solve puzzles, researchers now investigate why children make errors on mathematics and science problems.

Can educational psychology regain its position as the guiding science of education? Fortified with methods and theories that did not exist in Edward L. Thorndike's time, educational psychology can fulfill Wittrock's (1967) vision as the "scientific study of human behavior in educational settings" (p. 4). For example, Wittrock (1989) predicted that "recent advances in the study of cognition will lead to a useful and prescriptive theory of instruction and teaching" (p. 75).

Three Views of Learning and Instruction

Over the course of this century, the view of learning and instruction has changed in ways that have affected educational practice and research. In short, the following three metaphors of learning are relevant to instruction: learning as response acquisition, learning as knowledge acquisition, and learning as knowledge construction (Mayer, 1992a).

Learning as Response Acquisition

During the first half of this century, the learning as response acquisition metaphor dominated psychological theory and educational practice. Based on research on response learning in animals, learning was viewed as a mechanistic process in

¹ Cognitive theory refers to a body of concepts developed across several domains, including educational research, educational psychology, cognitive psychology, and cognitive science. Many of the cognitive issues originally examined within educational psychology have been studied more recently within cognitive psychology and cognitive science.

which successful responses are automatically strengthened and unsuccessful responses are automatically weakened, or more correctly, associations are strengthened or weakened according to environmental feedback. Within this behaviorist-inspired metaphor, the learner is a passive being whose repertoire of behaviors is determined by rewards and punishments encountered in the environment. Dewey (1902) disparagingly described this view of the passive learner: "The child is simply the immature being who is to be matured; he is the superficial being who is to be deepened. . . . It is his to receive, to accept" (p. 8).

The metaphor of learning as response acquisition has straightforward implications for instruction, namely, creating situations that elicit responses from learners and providing appropriate reinforcement for each response. Drill and practice is the epitome of instruction within this view of learning. For example, the student is given a response-eliciting question such as, "How do you spell *California*?" If the student produces the correct sequence of letters, *C-a-l-i-f-o-r-n-i-a*, the teacher smiles and says, "Right!" but if the student produces an incorrect sequence such as, "*C-o-w-i-l-l-f-o-u-r-n-y-a*," the teacher says, "You better look at the spelling list." If the learner is a passive recipient, then the teacher can be viewed as the active dispenser of feedback: someone who rewards correct responses and punishes incorrect ones.

According to this view of learning, the goal of instruction is to increase the number (or strength) of correct behaviors in the learner's repertoire. Learning outcomes can be evaluated by measuring the amount of behavior change, such as noting that a learner correctly spelled 5 of 12 words on Monday and correctly spelled 12 of 12 on Friday. By focusing on "how much is learned," the task of the teacher becomes one of shaping changes in specific responses. Although the behaviorist movement from which the metaphor of learning as response acquisition is derived has not fared well in the second half of this century, its remnants can be seen in modern theories of learning and instruction. For example, automatization of basic skills has become a component in modern theories of reading (LaBerge & Samuels, 1974), writing (Scardamalia, Bereiter, & Goelman, 1982), and mathematics (Fuson, 1982), as well as problem solving (Case, 1985; Mayer 1992b; Siegler, 1986) and skill learning (Anderson, 1983; Singley & Anderson, 1989).

Learning as Knowledge Acquisition

The cognitive revolution of the 1950s and 1960s emphasized a new metaphor: learning as knowledge acquisition. As the focus of research shifted from laboratory studies of animal learning to laboratory studies of human learning, psychologists shifted their theoretical focus from the strengthening/weakening of responses to the acquisition of knowledge. Within this new cognitive metaphor, the learner becomes a processor of information and the teacher becomes a dispenser of information. As the acquisition of knowledge becomes the focus of psychologists, the curriculum becomes the focus of instruction. Dewey (1902) described the curriculum-centered approach as follows:

Subdivide each topic into studies; each study into lessons; each lesson into specific facts and formulae. Let the child proceed step by step to master each one of these separate parts, and at last he will have covered the entire ground. (p. 8)

The learning as knowledge acquisition metaphor has straightforward implications for instruction, namely, in creating situations in which students can acquire knowledge. Through this metaphor, teaching basic information from textbooks and lectures becomes a focus of research and practice. According to this view of learning, the goal of instruction is to increase the amount of knowledge in the learner's repertoire so that learning outcomes can be evaluated by measuring the amount of knowledge acquired. For example, a multiple-choice reading comprehension test allows measurement of how much is learned, and the achievement test becomes the most popular evaluation device.

Learning as Knowledge Construction

As cognitive theory matured during the 1970s and 1980s, the dominant metaphor became learning as knowledge construction. Researchers moved from studying the learning of abstract materials in artificial settings to studying subject-based learning in more realistic situations. As a result, the view of the learner changed from that of a recipient of knowledge to that of a constructor of knowledge, an autonomous learner with metacognitive skills for controlling his or her cognitive processes during learning. Learning involves selecting relevant information and interpreting it through one's existing knowledge. As Resnick (1989) aptly noted, "learning occurs not by recording information but by interpreting it" (p. 2). Ninety years earlier, Dewey (1902) described this process of active learning: "Learning is active. It involves reaching out of one's mind. It involves organic assimilation from within" (p. 9). Accordingly, the teacher becomes a participant with the learner in the process of shared cognition, that is, in the process of constructing meaning in a given situation.

Concerning instruction, the focus changed from the curriculum to the cognition of the child. Dewey (1902) defined this child-centered approach as follows: "The child is the starting point, the center, the end. His development, his growth is the ideal" (p. 9). Thus, instruction is geared toward helping the student to develop learning and thinking strategies that are appropriate for working within various subject domains. Correspondingly, evaluation is qualitative rather than quantitative, determining how the student structures and processes knowledge rather than how much is learned.

Throughout this century, the dominant metaphor has changed from a view of the learner as a response-acquisition machine to a perception of the learner as a recipient of knowledge to a view of the learner as a constructor of knowledge. Although the third view appears to be new, it has had numerous proponents in the past (Bartlett, 1932; Tolman, 1932; Piaget, 1954). For this article, I adopt the learner as knowledge constructor metaphor in hopes of shedding light on the relation between cognition and instruction.

A Closer Look at Learners as Knowledge Constructors

In contrast to the classic behaviorist view of the learner as a passive recipient whose learning is automatically shaped by practice and reinforcements, the cognitive theory that has come to dominate psychology provides a vision of the learner as an active processor of information. In contrast to the behaviorist view of learning as the automatic strengthening or weakening of associations, cognitive theory offers the view of learning as the construction of knowledge by the learner. In contrast to the classic behaviorist focus on instruction that elicits individual responses that are immediately reinforced, a central instructional issue in cognitive theory concerns how to help students become more effective processors of information.

The main components in a cognitive model of the learning/instruction process (Mayer, 1987) include the following: *instructional manipulations*: the external events concerning what is taught and how it is taught; *learner characteristics*: the internal existing knowledge and information-processing system; *learning processes*: the internal cognitive processes engaged during learning; *learning outcomes*: the internal cognitive structures constructed during learning; and *outcome performance*: the external performance of the learner on tests.

This cognitive model includes internal factors that were ignored in behaviorist theory: learner characteristics (which can lead to individual differences in the process of learning), learning processes (which manipulate information), and learning outcomes (which contain the newly constructed knowledge).

Learning processes—information processes that affect the way in which new knowledge is constructed—provide an aspect of cognitive theory that is particularly relevant to education. For example, Figure 1 summarizes three cognitive learning processes involved in meaningful learning, which I

have proposed previously (Mayer, 1984, 1987, 1989b). According to the model, meaningful learning occurs when the learner selects relevant information, organizes that information into a coherent whole, and integrates that information with appropriate existing knowledge. In previous work I have called these three learning processes, respectively, *selecting*, *organizing*, and *integrating* (Mayer, 1984, 1987, 1989b, 1992a). These processes correspond respectively to Sternberg's (1985, 1988) three knowledge-acquisition components: selective encoding, selective combination, and selective comparison. Instructional manipulations are effective to the extent that they elicit cognitive learning processes.

The first process involves focusing attention on relevant pieces of the presented information and is indicated in Figure 1 by the arrow running from sensory memory to short-term memory. Selecting involves "selecting information from the text and adding that information to working memory" (Mayer, 1984, p. 32), and selective encoding involves a "sifting out of relevant from irrelevant information" (Sternberg, 1985, p. 107).

The second process requires building internal connections among (or organizing) the selected pieces of information, as indicated in Figure 1 by the arrow running from short-term memory back to short-term memory. Organizing involves "organizing the selected information in working memory into a coherent whole" (Mayer, 1984, p. 32), and selective combination involves "combining selectively encoded information in such a way as to form an integrated . . . internally connected whole" (Sternberg, 1985, p. 107).

The third process requires building external connections between the organized new knowledge and organized existing knowledge, as indicated in Figure 1 by the arrow directed from long-term memory to short-term memory. Integrating is "connecting the organized information to other familiar knowledge structures already in memory" (Mayer, 1984, p.

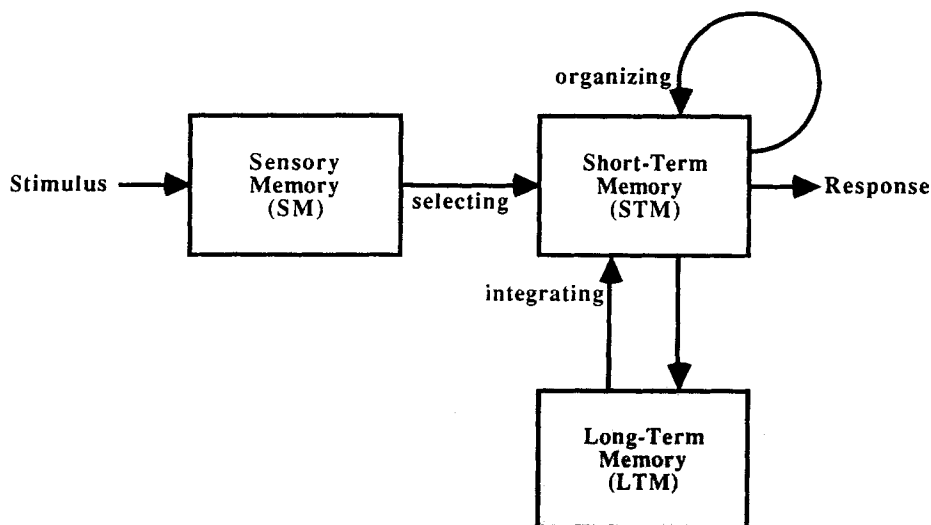


Figure 1. A cognitive model of knowledge construction. (From "Guiding Students' Cognitive Processing of Scientific Information," p. 246, by R. E. Mayer, 1992a, in M. Pressley, K. Harris, and J. Guthrie, Eds., *Promoting Academic Competence and Literacy: Cognitive Research and Instructional Innovation*, San Diego, CA: Academic Press. Copyright 1992 by Academic Press. Reprinted by permission.)

33), and selective comparison involves "relating newly acquired or retrieved information . . . to old knowledge so as to form an externally connected whole" (Sternberg, 1985, p. 107).

In cognitive theory, the learner is an active processor of information who is trying to make sense of the presented material. The self-regulated learner must appropriately control his or her learning processes by selecting and organizing relevant information and building connections from relevant existing knowledge. Accordingly, a primary goal of education is to help students develop expertise in how to learn and to use that expertise to construct useful knowledge within each subject domain. Although as educators we expect students to become expert learners, we rarely provide students with the opportunities they need to construct useful learning strategies (Mayer, 1987, 1992a; Norman, 1980; Pressley, 1990; Pressley & Levin, 1983a, 1983b; Weinstein & Mayer, 1985).

On the basis of current cognitive theory, Resnick (1989) has identified "three instructional aspects of learning that together call for forms of instructional theory very different from those that grew out of earlier associationist and behaviorist psychologies": (a) "learning is a process of knowledge construction, not knowledge recording or absorption," (b) "learning is knowledge-dependent" in that "people use current knowledge to construct new knowledge," and (c) "learning is highly tuned by the situation in which it takes place" (pp. 1-2). The constructive, knowledge-based, and situational aspects of learning have important implications for instructional research and practice, which are discussed in the next section.

Reemergence of Cognition and Instruction

With the reemergence of cognitive approaches to instruction now in full swing, it is fruitful to ask whether any new contributions have emerged and whether any are likely to emerge in the future. My reading of the educational psychology literature has convinced me that educational psychology has a unique and important role within the fields of psychology and education. In particular, I am struck by two contributions: a focus on cognition and instruction within subject matter domains (rather than on context-free general laws of learning and cognition) and a focus on detailed descriptions of individual differences in learning strategies and learning outcomes (rather than on average group differences).

First, in being driven by educational problems, educational psychologists are concerned with the ecological validity of their theories for real learning of real material in real situations. The development of psychologies of subject matter represents a unique contribution of educational psychology. For example, to paraphrase Resnick and Ford (1981), instead of asking, How do people learn, we ask, How do people learn to read and write and compute? Whereas a focus on general psychology failed to produce general laws of learning and cognition, domain-based research offers a potentially productive attempt to understand human learning within specific subject-matter domains.

Second, by focusing on the psychological processes and structures of individual learners, educational psychologists have opened a powerful line of study to further psychology's

goal of understanding what James (1899/1958) called the "mental machine . . . and its workings" (p. 26). The focus on learning strategies, including metacognitive strategies, and the focus on techniques for describing and evaluating knowledge structures represent unique contributions of educational psychology. Whereas measuring the average amount learned has not yielded a satisfactory psychology of learning, a more promising approach involves detailed evaluation of the learning processes and knowledge outcomes in individual learners.

In summary, two central aspects of the cognitive theory that have developed within educational psychology are a focus on learning and cognition within subject-matter domains and a focus on the detailed evaluation of individual learning strategies and outcomes. In the remainder of this section, I provide selected examples of the contribution of cognitive theory to the understanding of learning and instruction within the subject domains of reading, writing, and mathematics.

Reading

One of the major goals of reading instruction is reading comprehension, that is, the goal of helping readers to become effective processors of prose information. Cognitive theory emphasizes the constructive nature of reading comprehension and the central role of the reader's existing knowledge. The cognitive strategies used by successful readers include the following: determining importance, summarizing information, drawing inferences, generating questions, and monitoring comprehension (Dole, Duffy, Roehler, & Pearson, 1991). An implication, worthy of intensive study, is that instruction in reading comprehension should focus on helping readers to develop these useful cognitive strategies.

The strategy of determining importance involves distinguishing between important and unimportant information, such as distinguishing between the main idea and minor details. The strategy of summarizing involves taking the important information and organizing it, in one's own words, into a coherent structure. The strategy of drawing inferences involves filling in information omitted in a text to construct meaning for that text. For example, in hearing, "Our neighbor unlocked the door," a useful inference is that a key was used (Paris & Lindauer, 1976). The strategy of generating questions occurs when the reader asks specifically relevant questions about the text in advance of reading it and attempts to answer the questions while reading. Given the possibility that students may generate irrelevant questions, students appear to need specific training in how to generate useful questions (Pressley, 1990). The strategy of comprehension monitoring involves detecting that one does not understand a passage and correcting this lack of understanding, such as recognizing when there is a contradiction in a passage. Each of these strategies has been successfully taught to young readers in a way that has yielded improved performance on tests of reading comprehension, so each strategy is "a strong candidate for inclusion in our cognitively based comprehension curriculum" (Dole et al., 1991, p. 244).

In summary, cognitive theory has altered educators' views of how reading comprehension should be taught:

In rapid retreat is the view that comprehension ability consists of the independent sequential development of a set of hierarchically related skills, each learned to some level of mastery. . . . The development of reading comprehension ability is better viewed as a process of emerging expertise, where readers develop strategies for comprehending increasingly sophisticated texts. (Dole et al., 1991, p. 255).

Additional research is needed to determine how best to help students develop expert reading comprehension strategies.

Writing

In writing instruction students learn to express their ideas in written form. Cognitive theory emphasizes writing as communication rather than as composition, that is, the goal of writing is to communicate with an audience rather than solely to compose a technically correct product. The cognitive strategies used by successful writers include planning, translating, and reviewing (Flower & Hayes, 1980).

Planning involves generating information to be used in the essay, organizing that information into a writing plan, and establishing goals for writing. For example, to teach effective planning strategies, Fitzgerald and Teasley (1986) asked writers to begin by answering questions on a planning work sheet, such as "Who am I writing for?" "Why am I writing this?" "What do I know?" and "How can I group my ideas?"

Translating involves carrying out a plan by actually generating some written text. A major impediment for young writers is that the motoric (e.g., penmanship) and syntactic (e.g., spelling, punctuation, and grammar) demands of translating require so much mental effort that students do not pay attention to the semantic and pragmatic demands of writing, that is, making sure that the essay has the correct meaning and is appropriate for the intended audience. An instructional technique for improving the semantic and pragmatic aspects of translating is to remove the motoric and syntactic constraints on writing, at least during the first draft—that is, to allow students to orally dictate or to write in rough form without concern for spelling, punctuation, and grammar (Glynn, Britton, Muth, & Dogan, 1982).

The reviewing process involves detecting and correcting errors in written text. For example, Fitzgerald and Markman (1987) successfully taught young writers to conduct four types of revisions: additions, deletions, substitutions, and rearrangements. During instruction, teachers described their thought processes as they revised a text, asked students to describe their processes, and provided constructive feedback to students.

In short, Flower & Hayes's (1980) cognitive analysis of writing has led to an explosion of cognitively based writing programs that are beginning to be tested and refined. Cognitive theory has precipitated a fundamental shift from overemphasis on low-level skills (such as penmanship, spelling, and punctuation) to making room in the curriculum for the development of the processes central to writing: namely, planning, translating, and reviewing (Graham & Harris, 1988).

Mathematics

One of the major goals of mathematics instruction is the student's development of expertise in mathematical problem solving, such as the solving of word problems. Cognitive theory emphasizes the role of qualitative reasoning that enables successful mathematical problem solvers to understand the problem, devise and monitor a plan, and carry out the plan (Mayer, 1985, 1992b). An implication, worthy of intensive study, is that instruction in mathematical problem solving should focus on helping readers to develop these useful cognitive strategies.

Strategies for problem understanding involve translating each sentence from the problem statement into an internal mental representation and then organizing the relevant information into a coherent mental representation of the situation described in the problem statement (Kintsch & Greeno, 1985). To understand a problem, the problem solver must possess relevant factual knowledge (such as knowing that there are 12 in. in a foot), linguistic knowledge (such as knowing that "Joe has 5 more marbles than Pete" implies that "Pete has fewer marbles than Joe"), and schematic knowledge (such as knowing the difference between a river crossing problem and a work problem). The literature on the teaching of strategies for problem understanding is growing (Mayer, 1987, 1989a); for example, Lewis (1989) has successfully taught students how to translate sentences from a word problem into a diagrammatic representation.

Planning strategies involve breaking a problem into parts—that is, establishing subgoals—and determining which operations are needed to accomplish each subgoal. Monitoring strategies involve making sure that the plan is working and that the operations carried out are those called for in the plan. Can strategic knowledge, such as how to plan and monitor, be taught? There are encouraging signs that the answer is "Yes"; for example, Schoenfeld (1985) has shown that students can be taught to use planning strategies, such as breaking a problem into smaller parts.

Finally, the research on the automatization of arithmetic procedures shows that students sometimes possess faulty procedures. For example, Brown and Burton (1978) found that the error patterns of individual children in three-column subtraction could be accounted for by assuming that they correctly applied a subtraction procedure that contained one or more bugs, such as always subtracting the smaller number from the larger in a column. Detailed descriptions of students' procedural knowledge (including the bugs in faulty procedures) allow teachers to provide individualized corrective instruction for each student.

Cognitive theory represents a change in focus for mathematics education. Instead of asking, Which procedures should the student master? cognitive theorists ask What does it mean to think mathematically? (Schoenfeld, 1985). Instead of overemphasizing the product of problem solving (i.e., obtaining the right answer), cognitive theory suggests including an emphasis on the process of problem solving (i.e., what takes place in people's minds as they engage in solving complex mathematics problems).

Summary

In contrast to classical experimental psychology, which has sought general laws of learning and cognition, psychologies of subject matter show promise of making substantial contributions to educational practice. For example, by comparing the cognitive processes of successful and less successful students in domains such as reading comprehension, essay writing, or mathematical problem solving, psychologists of subject matter lay the groundwork for cognitively based instructional programs that help students develop domain-specific strategies.

In summary, psychologies of subject matter can provide the basis for educational reform. Recently, I described the role of psychologies of subject matter as follows:

Why have psychologies of subject matter flourished in the 1980s? Psychologists with an interest in human learning and cognition have become dissatisfied with studying solely contrived learning of artificial materials in laboratory settings; educators with an interest in subject matter have become unwilling to limit their studies to the nontheoretical issues of day-to-day classroom practices. The result is that the best of both worlds is emerging within the discipline of educational psychology: The methods and theories of cognitive psychology are being challenged by the educational problems of how students learn real subject matter. In short, this fruitful area of research called psychologies of subject matter is unique to the discipline of educational psychology. (Mayer, 1989a, p. 452)

The merger of cognitive and educational psychology that reenergized psychologies of subject matter during the 1980s offers a promising path to educational reform in the 1990s.

Conclusion

Cognitive theory offers a fundamentally new vision of what to teach, how to teach, and where to teach. The issue of what to teach involves focusing on abstract independent skills or concrete domain-based strategies; the issue of how to teach involves focusing on product (e.g., through drill and practice) or process (e.g., through modeling of underlying cognitive processes); and the issue of where to teach involves teaching in isolation or teaching within the context of a real problem or task. Instead of teaching individual behavioral skills in isolation through drill and practice (such as exercises in adding the suffix *s* to change a singular noun into a plural noun or adding a constant to both sides of an algebraic equation), cognitive theory suggests teaching of domain-based strategies within the context of larger problems through modeling of the required thinking processes (such as describing how one plans to write an essay on "How I Spent My Summer Vacation" or describing one's plan for solving a two-step word problem).

The years ahead promise to be exciting for those interested in fulfilling Thorndike's vision of improving educational practice on the basis of scientific research. The cognitive theory that has come to dominate psychology, and which was nurtured by educational demands, has finally begun to exert its influence on education. As the behaviorist grip on educational practices is replaced with cognitively oriented methods, the

time is ripe for fundamental changes in the process of instruction.

The educational reform movement, including the six national educational goals,² is replete with high-sounding proclamations from political leaders. However, educational psychologists have unique contributions to make to the success of this movement, so the educational decision makers would be missing a necessary component if they failed to base instructional changes on an understanding of how people learn, think, and develop. Wittrock (1989) eloquently summarized the central role of educational psychology as the meeting ground for cognition and instruction: "Educational psychology is the appropriate field for sustaining these close relations between cognitive psychologists and psychologists who focus on improving teaching and instruction" (p. 87).

² The six national educational goals are: "(1) By the year 2000, all children in America will start school ready to learn. (2) By the year 2000, the high school graduation rate will increase to at least 90 percent. (3) By the year 2000, American students will leave grades four, eight, and twelve having demonstrated competency over challenging subject matter including English, mathematics, science, history, geography, and every school in America will ensure that all students learn to use their minds well, so they may be prepared for responsible citizenship, further learning, and productive employment in our modern economy. (4) By the year 2000, American students will be first in the world in mathematics and science achievement. (5) By the year 2000, every adult American will be literate, and will possess the knowledge and skills necessary to compete in a global economy and exercise the rights and responsibilities of citizenship. (6) By the year 2000, every school in America will be free of drugs and violence and will offer a disciplined environment conducive to learning" (U.S. Department of Education, Office of Educational Research and Improvement, 1991, pp. 3-4).

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